
ALGEBRA 01

SINGLE CORRECT TYPE QUESTIONS

1. Let the system of linear equations

$$-x + 2y - 9z = 7$$

$$-x + 3y + 7z = 9$$

$$-2x + y + 5z = 8$$

$$-3x + y + 13z = \lambda$$

has a unique solution $x = \alpha$, $y = \beta$, $z = \gamma$. Then the distance of the point (α, β, γ) from the plane $2x - 2y + z = \lambda$ is :

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$$-x + 2y - 9z = 7$$

$$-x + 3y + 7z = 9$$

$$-2x + y + 5z = 8$$

$$-3x + y + 13z = \lambda$$

के अद्वितीय हल $x = \alpha$, $y = \beta$, $z = \gamma$ हो, तो बिन्दु (α, β, γ) की समतल $2x - 2y + z = \lambda$ से दूरी होगी :

(A) 13

(B) 9

(C) 11

(D) 7

Ans. D

Sol. $-x + 2y - 9z = 7$... (1)

$$-x + 3y + 7z = 9$$
 ... (2)

$$-2x + y + 5z = 8$$
 ... (3)

$$(B) - (1)$$

$$y + 16z = 2$$
 ... (4)

$$(C) - 2 \times (1)$$

$$-3y + 23z = -6$$
 ... (5)

$$3 \times (4) + (5)$$

$$71z = 0 \Rightarrow z = 0$$

$$y = 2$$

$$x = -3$$

$$(-3, 2, 0) \rightarrow (\alpha, \beta, \gamma)$$

$$\text{Put in } -3x + y + 13z = \lambda$$

$$\lambda = 9 + 2 = 11$$

$$d = \left| \frac{-6 - 4 - 11}{3} \right| = 7$$

2. Let A be a square matrix such that $AA^T = I$. Then $\frac{1}{2}A \left[(A + A^T)^2 + (A - A^T)^2 \right]$ is equal to :

माना एक वर्ग आव्यूह A के लिए $AA^T = I$ है। तो $\frac{1}{2}A \left[(A + A^T)^2 + (A - A^T)^2 \right]$ बराबर है :

- (A) $A^2 + I$ (B) $A^2 + A^T$ (C) $A^3 + I$ (D) $A^3 + A^T$

Ans. D

Sol. $AA^T = I = A^T A$

On solving given expression, we get

$$\begin{aligned} & \frac{1}{2}A \left[A^2 + (A^T)^2 + 2AA^T + A^2 + (A^T)^2 - 2AA^T \right] \\ &= A \left[A^2 + (A^T)^2 \right] = A^3 + A^T \end{aligned}$$

3. For $\alpha, \beta \in \mathbb{R}$ and a natural number n , let $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2} + \alpha \\ 2r & 2 & n^2 - \beta \\ 3r - 2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$. Then $2A_{10} - A_8$ is :

$\alpha, \beta \in \mathbb{R}$ और प्राकृतिक संख्या n के लिए, माना $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2} + \alpha \\ 2r & 2 & n^2 - \beta \\ 3r - 2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$, तो $2A_{10} - A_8$ है :

- (A) $4\alpha + 2\beta$ (B) $2n$ (C) 0 (D) $2\alpha + 4\beta$

Ans. A

Sol. $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2} + \alpha \\ 2r & 2 & n^2 - \beta \\ 3r - 2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$

$$R_2 \rightarrow R_2 - 2R_1$$

$$A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2} + \alpha \\ 0 & 0 & -\beta - 2\alpha \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$$

$$A_r = (\beta + 2\alpha)(3r - 3r + 2)$$

$$A_r = 4\alpha + 2\beta$$

$$\text{Hence } 2A_{10} - A_8$$

$$= A_{10} = 4\alpha + 2\alpha$$

4. Let A be a 3×3 real matrix such that $A^2(A - 2I) - 4(A - I) = O$, where I and O are the identity and null matrices, respectively. If $A^5 = \alpha A^2 + \beta A + \gamma I$, where α, β and γ are real constants, then $\alpha + \beta + \gamma$ is equal to:

माना 3×3 के एक वास्तविक आव्यूह A के लिए $A^2(A - 2I) - 4(A - I) = O$ है, जहाँ I और O क्रमशः तत्समक और शून्य आव्यूह हैं। यदि $A^5 = \alpha A^2 + \beta A + \gamma I$ है, जहाँ α, β और γ वास्तविक अचर हैं, तो $\alpha + \beta + \gamma$ बराबर है :

- (A) 76 (B) 20 (C) 12 (D) 4

Ans. C

Sol. $A^2(A - 2I) - 4(A - I) = 0$

$$A^3 - 2A^2 - 4A + 4I = 0$$

Multiply by A

$$A^4 = 2A^3 + 4A^2 - 4A$$

$$A^4 = 2(2A^2 + 4A - 4I) + 4A^2 - 4A$$

$$A^4 = 8A^2 + 4A - 8I$$

Multiply again by A

$$\Rightarrow A^5 = 8A^3 + 4A^2 - 8A$$

$$\Rightarrow A^5 = 8(2A^2 + 4A - 4I) + 4A^2 - 8A$$

$$\Rightarrow A^5 = 20A^2 + 24A - 32I$$

$$\text{Comparing with } A^5 = \alpha A^2 + \beta A + \gamma I$$

$$\alpha = 20, \beta = 24, \gamma = -32$$

$$\therefore \alpha + \beta + \gamma = 20 + 24 - 32$$

$$= 44 - 32$$

$$= 12$$

5. The values of α , for which
$$\begin{vmatrix} 1 & \frac{3}{2} & \alpha + \frac{3}{2} \\ 1 & \frac{1}{3} & \alpha + \frac{1}{3} \\ 2\alpha + 3 & 3\alpha + 1 & 0 \end{vmatrix} = 0$$
 lie in the interval :

α के वह मान, जिनके लिए
$$\begin{vmatrix} 1 & \frac{3}{2} & \alpha + \frac{3}{2} \\ 1 & \frac{1}{3} & \alpha + \frac{1}{3} \\ 2\alpha + 3 & 3\alpha + 1 & 0 \end{vmatrix} = 0$$
 है, किस अंतराल में है ?

(A) $(0, 3)$

(B) $\left(-\frac{3}{2}, \frac{3}{2}\right)$

(C) $(-2, 1)$

(D) $(-3, 0)$

Ans. D

Sol.
$$\begin{vmatrix} 1 & \frac{3}{2} & \alpha + \frac{3}{2} \\ 1 & \frac{1}{3} & \alpha + \frac{1}{3} \\ 2\alpha + 3 & 3\alpha + 1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow (2\alpha + 3) \left\{ \frac{7\alpha}{6} \right\} - (3\alpha + 1) \left\{ \frac{-7}{6} \right\} = 0$$

$$\Rightarrow (2\alpha + 3) \cdot \frac{7\alpha}{6} + (3\alpha + 1) \cdot \frac{7}{6} = 0$$

$$\Rightarrow 2\alpha^2 + 3\alpha + 3\alpha + 1 = 0$$

$$\Rightarrow 2\alpha^2 + 6\alpha + 1 = 0$$

$$\Rightarrow \alpha = \frac{-3 + \sqrt{7}}{2}, \frac{-3 - \sqrt{7}}{2}$$

6. Let the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ satisfy $A^n = A^{n-2} + A^2 - I$ for $n \geq 3$. Then the sum of all the elements of A^{50} is :

माना आव्यूह $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $A^n = A^{n-2} + A^2 - I$, $n \geq 3$ को संतुष्ट करता है। तो A^{50} के सभी अवयवों का योग है :

- (A) 39 (B) 52 (C) 53 (D) 44

Ans. C

Sol. $A^n - A^{n-2} = A^2 - I$

$$A^{50} - A^{48} = A^2 - I$$

$$A^{48} - A^{46} = A^2 - I$$

$\vdots$

$$A^4 - A^2 = A^2 - I$$

+

$$A^{50} - A^2 = 24(A^2 - I)$$

$$A^{50} = 25A^2 - 24I$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$$

7. Let $S = \left\{ x \in \mathbb{R} : (\sqrt{3} + \sqrt{2})^x + (\sqrt{3} - \sqrt{2})^x = 10 \right\}$. Then the number of elements in S is :

माना $S = \left\{ x \in \mathbb{R} : (\sqrt{3} + \sqrt{2})^x + (\sqrt{3} - \sqrt{2})^x = 10 \right\}$ है। तो S में अवयवों की संख्या है :

- (A) 4 (B) 0 (C) 1 (D) 2

Ans. D

Sol. $(\sqrt{3} + \sqrt{2})^x + (\sqrt{3} - \sqrt{2})^x = 10$

$$\text{Let } (\sqrt{3} + \sqrt{2})^x = t$$

$$t + \frac{1}{t} = 10$$

$$t^2 - 10t + 1 = 0$$

$$t = \frac{10 \pm \sqrt{100 - 4}}{2} = 5 \pm 2\sqrt{6}$$

$$(\sqrt{3} + \sqrt{2})^x = (\sqrt{3} \pm \sqrt{2})^2$$

$$x = 2 \text{ or } x = -2$$

Number of solutions = 2

8. Consider two sets A and B, each containing three numbers in A.P. Let the sum and the product of the elements of A be 36 and p respectively and the sum and the product of the elements of B be 36 and q respectively. Let d and D be the common differences of AP's in A and B respectively such that $D = d + 3, d > 0$. If $\frac{p+q}{p-q} = \frac{19}{5}$. then p - q is equal to :

दो समुच्चयों A और B का विचार कीजिए। दोनों में A.P. में तीन संख्याएँ हैं। माना A के अवयवों का योग और गुणनफल क्रमशः 36 और p हैं तथा B के अवयवों का योग और गुणनफल क्रमशः 36 और q हैं। माना A और B में AP's के सार्वअंतर क्रमशः d और D हैं। जिनके लिए $D = d + 3, d > 0$ हैं। यदि $\frac{p+q}{p-q} = \frac{19}{5}$ है, तो p - q बराबर है :

$$\text{लिए } D = d + 3, d > 0 \text{ हैं। यदि } \frac{p+q}{p-q} = \frac{19}{5} \text{ है, तो } p - q \text{ बराबर है :}$$

- (A) 600 (B) 540 (C) 450 (D) 630

Ans. B

Sol. $A = \{\alpha - d, \alpha, \alpha + d\} = \{12 - d, 12, 12 + d\}$

$$B = \{12 - D, 12, 12 + D\} = \{9 - d, 12, 15 + d\}$$

$$p = 12(144 - d^2)$$

$$q = 12(135 - 6d - d^2)$$

$$\frac{p+q}{p-q} = \frac{19}{5} \Rightarrow \frac{279 - 6d - 2d^2}{9 + 6d} = \frac{19}{5}$$

$$1395 - 30d - 10d^2 = 171 + 114d$$

$$10d^2 + 144d - 1224 = 0$$

$$5d^2 + 72d - 612 = 0$$

$$5d^2 + 102d - 30d - 612 = 0$$

$$\Rightarrow d = 6$$

$$D = 9$$

$$p - q = 540$$

9. Let $S_n = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \dots$ upto n terms. If the sum of the first six terms of an A.P. with first term $-p$ and common difference p is $\sqrt{2026S_{2025}}$, then the absolute difference between 20th and 15th terms of the A.P. is :

माना कि $S_n = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \dots$ n पदों तक है। यदि प्रथम पद $-p$ और सार्व अंतर p के साथ एक A.P. के पहले छः पदों का योग $\sqrt{2026S_{2025}}$ है, तो उस A.P. के 20^{वें} और 15^{वें} पदों के बीच का निरपेक्ष अंतर है :

- (A) 25 (B) 90 (C) 20 (D) 45

Ans. A

Sol. T_n of $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots$ is $\frac{1}{n^2 + n} = \frac{1}{n(n+1)}$

$$\begin{aligned} S_{2025} &= \sum_{n=1}^{2025} \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{2025} - \frac{1}{2026} \right) \\ &= 1 - \frac{1}{2026} = \frac{2025}{2026} \end{aligned}$$

Now,

$$\sqrt{2026 \times \frac{2025}{2026}} = \frac{6}{2} [-2p + (6-1)p]$$

$$45 = 3(3p)$$

$$p = 5$$

Now,

$$\begin{aligned} A_{20} - A_{15} &= [-5 + (19)(5)] - [-5 + (14)(5)] \\ &= 95 - 70 = 25 \end{aligned}$$

10. If the system of equations

$$x + (\sqrt{2} \sin \alpha)y + (\sqrt{2} \cos \alpha)z = 0$$

$$x + (\cos \alpha)y + (\sin \alpha)z = 0$$

$$x + (\sin \alpha)y - (\cos \alpha)z = 0$$

has a non-trivial solution, then $\alpha \in \left(0, \frac{\pi}{2}\right)$ is equal to :

यदि समीकरण निकाय

$$x + (\sqrt{2} \sin \alpha)y + (\sqrt{2} \cos \alpha)z = 0$$

$$x + (\cos \alpha)y + (\sin \alpha)z = 0$$

$$x + (\sin \alpha)y - (\cos \alpha)z = 0$$

का अतुच्छ हल है, तो $\alpha \in \left(0, \frac{\pi}{2}\right)$ बराबर है :

(A) $\frac{3\pi}{4}$

(B) $\frac{7\pi}{24}$

(C) $\frac{11\pi}{24}$

(D) $\frac{5\pi}{24}$

Ans. D

Sol.
$$\begin{vmatrix} 1 & \sqrt{2} \sin \alpha & \sqrt{2} \cos \alpha \\ 1 & \sin \alpha & -\cos \alpha \\ 1 & \cos \alpha & \sin \alpha \end{vmatrix} = 0$$

$$\Rightarrow 1 - \sqrt{2} \sin \alpha (\sin \alpha + \cos \alpha) + \sqrt{2} \cos \alpha (\cos \alpha - \sin \alpha) = 0$$

$$\Rightarrow 1 + \sqrt{2} \cos 2\alpha - \sqrt{2} \sin 2\alpha = 0$$

$$\cos 2\alpha - \sin 2\alpha = -\frac{1}{\sqrt{2}}$$

$$\cos \left(2\alpha + \frac{\pi}{4}\right) = -\frac{1}{2}$$

$$2\alpha + \frac{\pi}{4} = 2n\pi \pm \frac{2\pi}{3}$$

$$\alpha + \frac{\pi}{8} = n\pi \pm \frac{\pi}{3}$$

$$n = 0,$$

$$\alpha = \frac{\pi}{3} - \frac{\pi}{8} = \frac{5\pi}{24}$$

11. Let a and b be two distinct positive real numbers. Let 11th term of a GP, whose first term is a and third term is b, is

equal to p^{th} term of another GP, whose first term is a and fifth term is b . Then p is equal to :

माना a तथा b दो भिन्न धनात्मक वास्तविक संख्याएँ हैं। माना एक GP, जिसका पहला पद a तथा तीसरा पद b हैं, 11 वाँ पद, एक अन्य GP, जिसका पहला a तथा पाँचवा पद b है, के p वाँ पद के बराबर है। तो p बराबर है :

- (A) 21 (B) 20 (C) 25 (D) 24

Ans. A

Sol. G.P. first : $T_1 = a$

$$T_3 = b$$

$$\Rightarrow T_3 = ar^2 = b$$

$$\Rightarrow r = \pm \sqrt{\frac{b}{a}}$$

G.P. second : $T'_1 = a$

$$T'_5 = b$$

$$\Rightarrow T'_5 = ar_1^4 = b$$

$$\Rightarrow r_1 = \pm \left(\frac{b}{a}\right)^{\frac{1}{4}}$$

$$\Rightarrow T_{11} = ar^{10}$$

$$T'_p = ar_1^{p-1}$$

$$\Rightarrow T_{11} = T'_p \Rightarrow r^{10} = r_1^{p-1}$$

$$\Rightarrow \frac{b^5}{a^5} = \left(\frac{b}{a}\right)^{\frac{p-1}{4}}$$

$$\Rightarrow \frac{p-1}{4} = 5 \quad \Rightarrow p = 21$$

12. Let T_r be the r^{th} term of an A.P. If for some m , $T_m = \frac{1}{25}$, $T_{25} = \frac{1}{20}$, and $20 \sum_{r=1}^{25} T_r = 13$, then $5m \sum_{r=m}^{2m} T_r$ is equal to :

माना कि T_r एक A.P. का $r^{\text{वाँ}}$ पद है। किसी m के लिए, $T_m = \frac{1}{25}$, $T_{25} = \frac{1}{20}$, और $20 \sum_{r=1}^{25} T_r = 13$ है, तो $5m \sum_{r=m}^{2m} T_r$ बराबर है :

(A) 126

(B) 142

(C) 98

(D) 112

Ans. A

Sol. $T_m = \frac{1}{25}, T_{25} = \frac{1}{20}, 20 \sum_{r=1}^{25} T_r = 13$

$$T_m = a + (m-1)d = \frac{1}{25} \quad \dots(1)$$

$$T_{25} = a + 24d = \frac{1}{20}$$

$$20 \cdot \frac{25}{2} \left[a + \frac{1}{20} \right] = 13 \Rightarrow a = \frac{1}{500}$$

$$\text{also, } 20S_{25} = 20 \cdot \frac{25}{2} [2a + 24d] = 13 \Rightarrow d = \frac{1}{500}$$

$$\text{from (1)} \quad \frac{1}{500} + \frac{m-1}{500} = \frac{1}{25} \Rightarrow m = 20$$

Now,

$$5m \sum_{r=m}^{2m} T_r = 100 \sum_{r=20}^{40} T_r = 126$$

13. Let α, β be the roots of the equation $x^2 + 2\sqrt{2}x - 1 = 0$. Then quadratic equation, whose roots are $\alpha^4 + \beta^4$ and $\frac{1}{10}(\alpha^6 + \beta^6)$, is :

माना समीकरण $x^2 + 2\sqrt{2}x - 1 = 0$ के मूल α, β हैं। द्विघात समीकरण, जिसके मूल $\alpha^4 + \beta^4$ तथा $\frac{1}{10}(\alpha^6 + \beta^6)$ हैं, है :

(A) $x^2 - 180x + 9506 = 0$

(B) $x^2 - 195x + 9466 = 0$

(C) $x^2 - 195x + 9506 = 0$

(D) $x^2 - 190x + 9466 = 0$

Ans. C

Sol. $x^2 + 2\sqrt{2}x - 1 = 0$

$$\alpha + \beta = -2\sqrt{2}$$

$$\alpha\beta = -1$$

$$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$$

$$= ((\alpha + \beta)^2 - 2\alpha\beta)^2 - (\alpha\beta)^2$$

$$= (8 + 2)^2 - 2(-1)^2$$

$$= 100 - 2 = 98$$

$$\alpha^6 + \beta^6 = ((\alpha + \beta) ((\alpha + \beta)^2 - 3\alpha\beta))^2 - 2(\alpha\beta)^3$$

$$= (-2\sqrt{2}(8+3))^2 + 2$$

$$= (8)(121) + 2 = 970$$

$$\frac{1}{10}(\alpha^6 + \beta^6) = 97$$

$$x^2 - (98 + 97)x + (98)(97) = 0$$

$$\Rightarrow x^2 - 195x + 9506 = 0$$

14. The number of solutions of equation $\left(\frac{9}{x} - \frac{9}{\sqrt{x}} + 2\right)\left(\frac{2}{x} - \frac{7}{\sqrt{x}} + 3\right) = 0$ is :

समीकरण $\left(\frac{9}{x} - \frac{9}{\sqrt{x}} + 2\right)\left(\frac{2}{x} - \frac{7}{\sqrt{x}} + 3\right) = 0$ के हलों की संख्या है :

(A) 1

(B) 4

(C) 2

(D) 3

Ans. B

Sol. $\left(\frac{9}{x} - \frac{9}{\sqrt{x}} + 2\right)\left(\frac{2}{x} - \frac{7}{\sqrt{x}} + 3\right) = 0$

$$\frac{9}{x} - \frac{9}{\sqrt{x}} + 2 = 0$$

or

$$\frac{2}{x} - \frac{7}{\sqrt{x}} + 3 = 0$$

$$\sqrt{x} = t$$

$$\sqrt{x} = u$$

$$x = t^2$$

$$\frac{9}{t^2} - \frac{9}{t} + 2 = 0$$

$$\frac{2}{u^2} - \frac{7}{u} + 3 = 0$$

$$9 - 9t + 2t^2 = 0$$

$$2 - 7u + 3u^2 = 0$$

$$2t^2 - 9t + 9 = 0$$

$$3u^2 - 7u + 2 = 0$$

$$2t^2 - 6t - 3t + 9 = 0$$

$$3u^2 - 6u - u + 2 = 0$$

$$2t(t - 3) - 3(t - 3) = 0$$

$$3u(u - 2) - 1(u - 2) = 0$$

$$t = 3 \text{ or } t = \frac{3}{2}$$

$$u = 2 \text{ or } u = \frac{1}{3}$$

$$\sqrt{x} = 3, \sqrt{x} = \frac{3}{2}$$

$$\sqrt{x} = 2, \sqrt{x} = \frac{1}{3}$$

$$x = 9, x = \frac{9}{4}$$

$$x = 4, x = \frac{1}{9}$$

15. Consider the equation $x^2 + 4x - n = 0$, where $n \in [20, 100]$ is a natural number. Then the number of all distinct values of n , for which the given equation has integral roots, is equal to :

समीकरण $x^2 + 4x - n = 0$, जहाँ $n \in [20, 100]$ एक धनपूर्णांक है, का विचार कीजिए। तो n के सभी भिन्न मानों, जिनके लिए दी गई समीकरण के मूल पूर्णांक हैं, की संख्या है :

- (A) 6 (B) 8 (C) 5 (D) 7

Ans. A

Sol. $x^2 + 4x - n = 0$

$$n \in [20, 100]$$

Equation has integral roots

$$x = \frac{-4 \pm \sqrt{16 + 4n}}{2}$$

$$= \frac{-4 \pm 2\sqrt{4 + n}}{2}$$

$$= -2 \pm \sqrt{4 + n}$$

For integral roots $8 + 2n$ must be a complete square.

Now, $4 + n$ can range from $[24, 104]$

Complete squares in this range 25, 36, 49, 64, 81, 100

$\therefore$ Total possible values = 6

16. Let $\alpha\beta \neq 0$ and $A = \begin{bmatrix} \beta & \alpha & 3 \\ \alpha & \alpha & \beta \\ -\beta & \alpha & 2\alpha \end{bmatrix}$. If $B = \begin{bmatrix} 3\alpha & -9 & 3\alpha \\ -\alpha & 7 & -2\alpha \\ -2\alpha & 5 & -2\beta \end{bmatrix}$ is the matrix of cofactors of the elements of A ,

then $\det(AB)$ is equal to :

माना $\alpha\beta \neq 0$ तथा $A = \begin{bmatrix} \beta & \alpha & 3 \\ \alpha & \alpha & \beta \\ -\beta & \alpha & 2\alpha \end{bmatrix}$ हैं। यदि आव्यूह A के अवयवों के सहखंडों का आव्यूह $B = \begin{bmatrix} 3\alpha & -9 & 3\alpha \\ -\alpha & 7 & -2\alpha \\ -2\alpha & 5 & -2\beta \end{bmatrix}$ है, तो

$\det(AB)$ बराबर है :

- (A) 125 (B) 64 (C) 343 (D) 216

Ans. D

Sol. cofactor of 1-1 element = $2\alpha^2 - \alpha\beta = 3\alpha$

$$\Rightarrow 2\alpha - \beta = 3$$

Cofactor of 2-1 element = $-(2\alpha^2 - 3\alpha) = -\alpha$

$$\Rightarrow 2\alpha - 3 = 1 \Rightarrow \alpha = 2$$

$$\Rightarrow \beta = 1 \quad \Rightarrow |A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 2 & 1 \\ -1 & 2 & 4 \end{vmatrix} = 6$$

17. Let $\alpha, \beta; \alpha > \beta$, be the roots of the equation $x^2 - \sqrt{2}x - \sqrt{3} = 0$. Let $P_n = \alpha^n - \beta^n, n \in \mathbb{N}$. Then $(11\sqrt{3} - 10\sqrt{2})P_{10} + (11\sqrt{2} + 10)P_{11} - 11P_{12}$ is equal to :

माना समीकरण $x^2 - \sqrt{2}x - \sqrt{3} = 0$ के मूल $\alpha, \beta; \alpha > \beta$ हैं। माना $P_n = \alpha^n - \beta^n, n \in \mathbb{N}$ है। तो $(11\sqrt{3} - 10\sqrt{2})P_{10} + (11\sqrt{2} + 10)P_{11} - 11P_{12}$ बराबर है :

- (A) $11\sqrt{2}P_9$ (B) $11\sqrt{3}P_9$ (C) $10\sqrt{3}P_9$ (D) $10\sqrt{2}P_9$

Ans. C

Sol. $x^2 - \sqrt{2}x - \sqrt{3} = 0$

$$P_n = \alpha^n - \beta^n$$

α and β are the roots of the equation

Using Newton's theorem

$$P_{n+2} - \sqrt{2}P_{n+1} - \sqrt{3}P_n = 0$$

Put $n = 10$

$$P_{12} - \sqrt{2}P_{11} - \sqrt{3}P_{10} = 0$$

$$P_{12} = \sqrt{2}P_{11} + \sqrt{3}P_{10}$$

Put $n = 9$

$$P_{11} - \sqrt{2}P_{10} - \sqrt{3}P_9 = 0$$

$$P_{11} = \sqrt{2}P_{10} + \sqrt{3}P_9$$

$$(11\sqrt{3} - 10\sqrt{2})P_{10} + (11\sqrt{2} + 10)P_{11} - 11P_{12}$$

Put the value of P_{12} and P_{11} in above equation.

$$\begin{aligned}
 &= (11\sqrt{3} - 10\sqrt{2})P_{10} + (11\sqrt{2} + 10)(\sqrt{2}P_{10} + \sqrt{3}P_9) - 11(\sqrt{2}P_{11} + \sqrt{3}P_{10}) \\
 &= 11\sqrt{3}P_{10} - 10\sqrt{2}P_{10} + 22P_{10} + 10\sqrt{2}P_{10} + 11\sqrt{6}P_9 + 10\sqrt{3}P_9 - 11\sqrt{2}P_{11} - 11\sqrt{3}P_{10} \\
 &= 22P_{10} + 11\sqrt{6}P_9 + 10\sqrt{3}P_9 - 11\sqrt{2}(\sqrt{2}P_{10} + \sqrt{3}P_9) \\
 &= 22P_{10} + 11\sqrt{6}P_9 + 10\sqrt{3}P_9 - 22P_{10} - 11\sqrt{6}P_9 \\
 &= 10\sqrt{3}P_9
 \end{aligned}$$

18. If each term of a geometric progression $a_1, a_2, a_3, \dots$ with $a_1 = \frac{1}{8}$ and $a_2 \neq a_1$, is the arithmetic mean of the next two terms and $S_n = a_1 + a_2 + \dots + a_n$, then $S_{20} - S_{18}$ is equal to :

यदि गुणोत्तर श्रेणी $a_1, a_2, a_3, \dots$ जिसमें $a_1 = \frac{1}{8}$ तथा $a_2 \neq a_1$ अगले दो पदों का समांतर माध्य है तथा

$S_n = a_1 + a_2 + \dots + a_n$ है, तो $S_{20} - S_{18}$ बराबर है :

- (A) 2^{15} (B) 2^{18} (C) -2^{15} (D) -2^{18}

Ans. C

Sol. $a_2 = \frac{a_3 + a_4}{2}$

$$\Rightarrow 2ar = ar^2 + ar^3$$

$$\Rightarrow r^2 + r = 2$$

$$\Rightarrow r = 1 \text{ or } -2$$

$$\Rightarrow r \neq 1 \because (a_1 \neq a_2)$$

$$\Rightarrow r = -2$$

$$\text{Now } S_{20} - S_{18} = T_{19} + T_{20}$$

$$= ar^{18} + ar^{19}$$

$$= ar^{18}(1 + r)$$

$$\frac{1}{8}(-2)^{18}(-1) = -2^{15}$$

19. Let $a_1, a_2, a_3, \dots$ be in an A.P. such that $\sum_{k=1}^{12} a_{2k-1} = -\frac{72}{5}a_1, a_1 \neq 0$. If $\sum_{k=1}^n a_k = 0$, then n is :

माना $a_1, a_2, a_3, \dots$ एक A.P. में हैं और $\sum_{k=1}^{12} a_{2k-1} = -\frac{72}{5}a_1, a_1 \neq 0$ हैं। यदि $\sum_{k=1}^n a_k = 0$ है, तो n है :

(A) 17

(B) 11

(C) 10

(D) 18

Ans. B

Sol. Let $a_1 = a$, common difference = d

$$a_1 + a_3 + a_5 + \dots + a_{23} = -\frac{72}{5}a$$

$$\frac{12}{2}[2a + 11 \times 2d] = -\frac{72}{5}a$$

$$12a + 132d = -\frac{72}{5}a$$

$$132a + 132 \times 5d = 0$$

$$a = -5d$$

$$\frac{n}{2}(2a + (n-1)d) = 0 \Rightarrow -10d + nd - d = 0$$

$$n = 11$$

20. Let x_1, x_2, x_3, x_4 be in a geometric progression. If 2, 7, 9, 5 are subtracted respectively from x_1, x_2, x_3, x_4 , then the resulting numbers are in an arithmetic progression. Then the value of $\frac{1}{24}(x_1 x_2 x_3 x_4)$ is :

माना x_1, x_2, x_3, x_4 एक गुणोत्तर श्रेणी में हैं। यदि x_1, x_2, x_3, x_4 में से क्रमशः 2, 7, 9, 5 घटाने पर प्राप्त संख्याएँ एक समान्तर श्रेणी में हैं, तो $\frac{1}{24}(x_1 x_2 x_3 x_4)$ का मान है :

(A) 216

(B) 18

(C) 72

(D) 36

Ans. A

Sol. x_1 x_2 x_3 x_4

$$a \quad ar \quad ar^2 \quad ar^3$$

$$a - 2, \quad ar - 7, \quad ar^2 - 9, \quad ar^3 - 5 \quad \rightarrow \text{A.P.}$$

$$2(ar - 7) = a - 2 + ar^2 - 9 \Rightarrow ar^2 - 2ar + a = -14 + 11 \quad \dots(1)$$

$$2(ar^2 - 9) = ar - 7 + ar^3 - 5 \Rightarrow ar^3 - 2ar^2 + ar = -18 + 12 \quad \dots(2)$$

$$\frac{(2)}{(1)} = \frac{r^3 - 2r^2 + r}{r^2 - 2r + 1} = \frac{-6}{-3} \Rightarrow r = 2 \text{ \& } a = -3$$

$$\text{So } \frac{1}{24}x_1 x_2 x_3 x_4 = \frac{1}{24} \times a^4 \cdot r^6 = \frac{1}{24} \times 81 \times 64 = 216$$